

The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. End-to-end crypto quant for public LLMs needs a reliable *data end* before decision or execution: models fail when they cannot see a fragmented, asynchronous market. We contribute that data end as a crypto *observation layer*—an anonymized world-model runtime (state compiler + quality gate, not a generative simulator)—that turns multi-band evidence into a point-in-time (PIT) *world bundle* AI consumers can read. Hard refusal is the data-end safety valve: when support is thin, a machine-readable flag blocks action so half-baked context cannot drive downstream decisions. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. ≤ 0.23 under Raw. Secondary—valve: across four live vendors (temperature 0), Compiled thin-abstain is 1.0 while Ungated soft judgment on the *same* content is only ≈ 0.68 ; Raw over-trades and loses; Blind refuses with no feed. Secondary—content: a no-LLM rule on vintaged tilts beats momentum ($\Delta CE=0.334$, $p=0.034$); on the open slice ($WMI \geq 0.05$; $n=1224$) Ungated CE is positive for every vendor (four-vendor mean $\approx +0.29$; bootstrap CIs include 0) as a field lower bound for the compiled view. The panel is scarce (max $WMI \approx 0.093 < 0.2$), so production Compiled never opens—a valve stress test of the data end. **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Build the *data end* of end-to-end crypto quant for public LLMs—the observation layer that turns asynchronous market evidence into a consumable world. End-to-end quant needs a stack: **data/observation** → **decision** → **execution**. Finance agents [19–21] and generative world models [6, 7, 9] typically own decision (and sometimes execution) while *assuming* a usable observation stream. Crypto violates that assumption: exchange microstructure, macro vintages, and alternative flows arrive on incompatible clocks [11, 13]. This paper owns only the data end: compile evidence into a PIT world bundle so models can see; attach

a hard quality valve so half-baked context cannot drive downstream decisions. Refusal is the data-end safety valve—not a substitute for seeing, and not a trading product.

Failure modes we target. Without a feed, public models refuse (safe but useless as a data end). With a thin price fragment, they often trade and lose (unsafe for any downstream stack). With a disclosed but ungated world, they read fields yet disagree on whether to act. The data end must therefore (i) make the market *legible* and (ii) attach an enforceable quality valve before decision/execution.

What we build. An anonymized *world-model runtime* (Section 4) is the data-end compiler: multi-band evidence → BandPIT/WMI → JSON *world bundle* that GPT /DeepSeek /GLM /Gemini-class models can read. “World model” here means observation compiler + quality-gated runtime (Section 3), not next-state dynamics and not a decision engine. The bundle is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence IDs)—that package is the market view handed to the next stage. WMI/ACWMI and `should_ai_abstain` attach a machine-readable valve when the view is too thin to act.

How we evaluate. **RQ1 (primary—seeing):** Does the Compiled data end make ready /missing /tilt answers verifiable against PIT ground truth? **RQ2 (safety valve):** When the world is thin, does a hard flag enforce abstention where soft disclosure does not? Four arms on identical days: Compiled (bundle + hard flag), Ungated (same content, no flag), Raw (mom5), Blind (no feed). **RQ3 (secondary):** Are vintaged tilts economically nonempty under a transparent no-LLM rule—a content check on the compiled view, not a full decision-layer claim?

Contributions. (1) Data-end semantics: epistemic observations, compilation Π_t , and a clean split between *seeing* (bundle) and *gating* (hard flag) at the observation layer of the quant stack. (2) Anonymized PIT runtime: vintage collectors, BandPIT previous-close clock, quality-tagged bundles, frozen four-arm harness—the compile path of the data end. (3) Live + rule validation on a scarce three-band archive: grounding ≈ 0.97 vs. Raw collapse; hard-gate thin-abstain 1.0 vs. Ungated ≈ 0.68 ; nonempty tilts ($\Delta CE=0.334$) and an open-slice content lower bound. Estimand: data-end legibility and valve reliability—not decision-layer SOTA, not execution, and not production Compiled-open alpha.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how

agents fetch, debate, and are scored. They still assume a usable observation stream—the data end of the quant stack. We study that prior stage: compile a world bundle, measure whether models can see it, then gate action when they must not act.

Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to world quality via a runtime field, not classifier confidence alone. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation Layer (Data End)

The data end of the stack is an observation layer: compile raw evidence into a world the model can read, then attach a quality valve.

Definition 3.1 (Epistemic observation). $O_{j,t} = (x, \tau, q, g, r)$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (WMI/ACWMI); $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$. Compilation Π_t aligns clocks, applies honesty/missingness gates, and assembles band views into consumer JSON—the handoff object to decision.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $WMI_t = B_t U_t H_t$ aggregates breadth, stability, and honesty; ACWMI supports regime-conditional gating. The Compiled data end exposes `should_ai_abstain` and `thin_world` as first-class fields.

Seeing vs. gating. The bundle is what the model *sees* (completeness, tilts, evidence IDs). The hard flag is how the data end *blocks* action when $q(W_t)$ is too low. Tilts without gates are still a dump; a gate without a readable bundle is only a switch.

PROPOSITION 3.3 (COMPILATION $\neq$ DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty.*

PROPOSITION 3.4 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

REMARK 3.5 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface.*

4 Runtime: The Data-End Compile Path

The system object is the data end of the stack: an observation layer whose job is to make the market *visible* to public LLMs before decision or execution. Figure 1 and Table 1 summarize the anonymized prototype (paths omitted for double-blind review): vintage collectors write multi-band history; BandPIT compiles a PIT world state; a service emits a JSON *world bundle*; frozen-prompt consumers read that bundle at temperature 0. Decision and execution sit *downstream* of this compile path; this paper does not claim them.

Table 1: Data-end modules that build the view (not a full product inventory).

Module	Role in the data end
Collectors / stores	Ingest timestamped / vintaged band history
BandPIT compiler	Align clocks; grade ready/stale/missing; Π_t
Quality indices	$B, U, H \rightarrow$ WMI; abstain flag
Bundle builder	Emit complete / honest / auditable JSON
Consumer harness	Four-arm LLM protocol; temp. 0

Table 2: Input evidence vs. fields emitted in the world bundle.

Band	Fetchd (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, abstain flag, evidence IDs

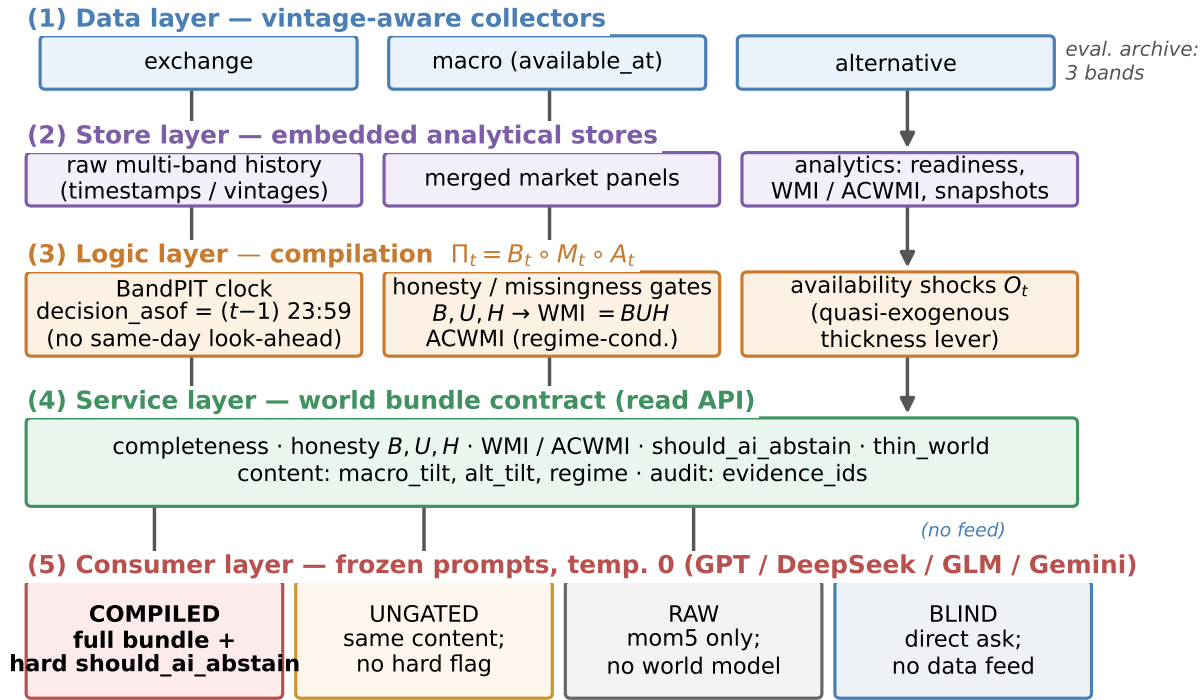
What enters the view. This paper evaluates three bands that populate the consumer object (Table 2). *Exchange*: OHLCV klines, funding, open interest. *Macro*: equity-volatility and dollar indices with `available_at` vintages. *Alternative*: aggregate stablecoin circulation. Every row carries an observation timestamp so compilation can enforce a previous-close information set. The schema is band-extensible; the experiments below score the view built from these three feeds.

How compilation builds the view. For calendar day t , BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return). Each band is graded ready / limited / missing against age thresholds; missingness is disclosed rather than imputed. Breadth, stability, and honesty aggregate into $WMI_t = B_t U_t H_t$ (ACWMI adds a regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The valve sets `should_ai_abstain` when $WMI < 0.2$ and exposes `thin_world`—a switch on the view, not a substitute for seeing. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days.

What the LLM sees: the world bundle. The consumer object is a JSON world bundle (Table 3, Listing 1): timing, completeness, quality (including the abstain flag), content tilts, and evidence IDs that bind rationales to disclosed bands. That package is the market view we ask public LLMs to read. Outputs used in experiments are a $\sim 3,990$ -row PIT panel (10 assets $\times$ 399 days; 2025-07 to 2026-08) and the per-day bundles fed to the four arms.

Listing 1: World bundle example (what Compiled sees).

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1
  },
  "world_model_index": {
    "wmi": 0.015,
    "should_ai_abstain": true,
    "thin_world": true
  },
  "macro_tilt": 1.0,
  "alt_tilt": -1.0,
  "audit": {
    "evidence_ids": [
      "band:exchange:missing",
      "band:macro:ready"
    ]
  }
}
```



Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).

Figure 1: Data-end pipeline: raw multi-band evidence → BandPIT compilation → world bundle (what the LLM sees) → Compiled / Ungated / Raw / Blind consumers.

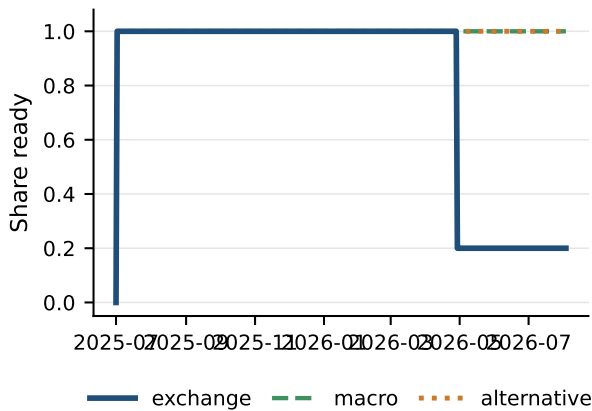


Figure 2: PIT readiness: what is visible in the archive over time.

How consumers use the view. **Compiled:** full bundle + hard abstain when the quality flag is true (prompt also forbids acting

Table 3: World-bundle field groups (the market view).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	ready/limited/missing counts, band status
Quality	WMI, ACWMI, abstain flag, thin_world
Content	macro_tilt, alt_tilt, mom5
Audit	evidence_ids

when the world is judged thin). **Ungated:** same content and numeric WMI; soft “you may abstain.” **Raw:** mom5 only—a thin fragment without the compiled view. **Blind:** no feed. On this panel max WMI ≈ 0.093 , so the production gate marks every OOS day thin and Compiled abstains throughout—stressing the valve while RQ1 still scores whether the disclosed view is legible.

Frozen prompts (gating only). Both Compiled and Ungated consume the same bundle content; only enforcement differs (Listings 2–3). Raw omits world-quality language. Credentials stay in environment variables; prompts, IS/OOS cut, and seeds are frozen for review.

Listing 2: Compiled (excerpt): hard valve.

```
1. If should_ai_abstain is true OR the world is thin,
```

Table 4: Grounding (50 days): primary seeing result. C = Compiled bundle; R = Raw.

Model	Ready F1		Missing F1		Tilt-sign	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

```

you MUST "abstain".
2. Prefer band content over raw momentum.
3. Rationale must match evidence_ids.

```

Listing 3: Ungated (excerpt): soft judgment.

```

1. Judge whether low WMI / missingness makes action
   unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.

```

5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, chronological IS/OOS 200/200 days. Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors. The no-LLM probe uses 199 OOS days after return alignment. Primary metrics for RQ1 are grounding F1/accuracy; for RQ2, thin-world abstain rate (CE as avoided loss, not alpha).

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* view of the market—not invent coverage from a price fragment, and not confuse “bands present” with “world thick enough to trade.”

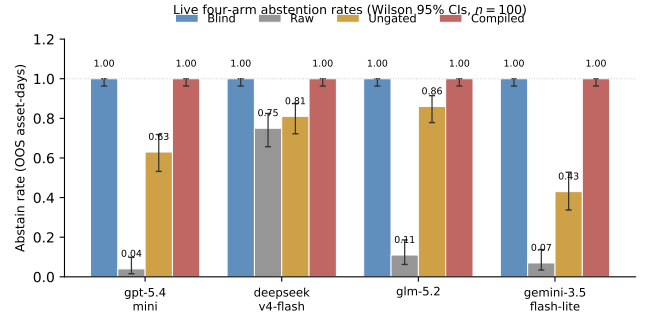
On 50 OOS asset-days, given Compiled or Raw inputs, each model must (i) list ready bands among exchange/macro/alternative, (ii) list missing bands, (iii) report signs of macro_tilt and alt_tilt, and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth.

Table 4: Compiled mean ready F1 0.97, missing F1 0.97, tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. This is not a trivial copy test in the product sense: Raw has no readiness or tilt fields, so models cannot recover the archive state; Compiled forces the consumer to *use* the disclosed world rather than a momentum guess. Sufficiency chat alone is weak—models may call a 3/3-ready day “enough” while WMI still marks the world thin—so we separate legibility (RQ1) from the hard valve (RQ2).

Case study (same asset-day). Listings 4–5 excerpt live rationales on ADA (2026-05-11): exchange missing; macro/alternative ready; thin world. Compiled recovers the checkable view and judges the world insufficient. Raw cannot list readiness or tilt signs because the fragment has no world fields. A second pair on ADA (2026-01-17) shows the valve contrast: Compiled abstains because `should_ai_abstain` is true; Raw goes bullish from `mom5` = +1 alone—half-baked context driving action.

Table 5: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

**Figure 3: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.****Listing 4: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).**

```

ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"

```

Listing 5: Raw grounding / action (same model family).

```

grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."

```

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. Table 5 and Figure 3: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 6); Blind abstains 1.0 with no feed. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean ΔCE (Compiled–Raw) +1.07 is avoided loss under the valve, not an alpha claim. Full-OOS ($n=2000$, all thin) reproduces Compiled 1.0 and Ungated mean ≈ 0.56 .

Open slice under disclosure. Table 7: the production WMI gate 0.2 marks every OOS day thin on this panel (max WMI ≈ 0.093); a 0.05 cut selects the band-thick subset ($\approx 61\%$ of OOS). Table 8: on that slice Ungated keeps compiled content without the hard boolean—mean abs. ≈ 0.48 , CE positive for every vendor (four-vendor mean $\approx +0.29$; bootstrap mean $\approx +0.31$). Production Compiled still abstains at 1.0 by the 0.2 gate; Raw remains risky (GPT/Gemini abstain 0.0,

Table 6: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 7: WMI gate vs. open share (OOS). Production default never opens on this panel.

Gate	Open share	Note
0.2 (production)	0.00	max WMI ≈ 0.093
0.05 (counterfactual)	0.61	$\equiv$ band-thick
0.01	1.00	near-floor open

Table 8: Open slice (3/3 ready $\equiv$ WMI ≥ 0.05 ; $n \approx 1224$). Content lower bound, not Compiled-open.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	-0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	-0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	-0.25

*Raw GLM $n=398$; Ungated GLM $n=932$; others $n=1224$. Bootstrap mean CE $\approx +0.31$ ($B=5,999$ reps; one-sided $p \approx 0.21$).

Table 9: No-LLM content probe (secondary).

Policy	Sharpe	CE
Buy-and-hold	-1.404	-1.163
Momentum	0.096	-0.206
World-bundle tilts	0.764	+0.130
Tilts – Momentum	—	0.334 ($p=0.034$)

CE negative). The Ungated numbers bound the economic content of disclosed fields under soft judgment.

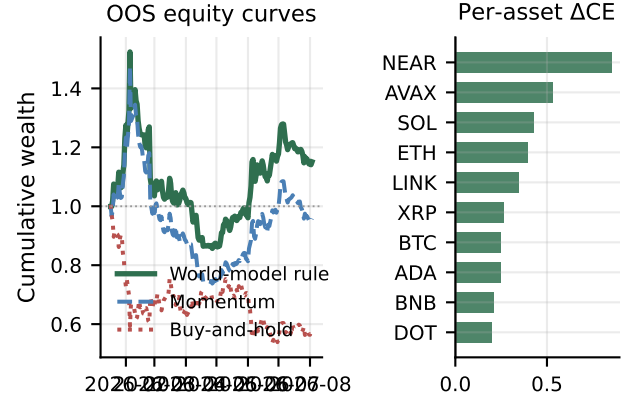
5.4 RQ3 (secondary): Nonempty bundle fields

Seeing and gating do not require the tilts to be economically empty. A transparent no-LLM rule adds vintaged `macro_tilt` / `alt_tilt` to return inputs shared with momentum (Table 9, Figure 4). OOS (199 days after return alignment $\times 10$ assets): Sharpe 0.764 / CE +0.130 vs. momentum (0.096, -0.206) and buy-and-hold (-1.404, -1.163); block-bootstrap $\Delta CE = 0.334$ ($p=0.034$). Leave-one-band-out attributes the gap to macro/alternative *content* (content share 1.0 under an ungated probe), not to the refuse flag. A long non-vintaged audit shows no edge: value appears where the vintage-compiled world exists. Table 10: on the open/thick slice the content rule beats momentum ($\Delta CE = 0.315$); on exchange-gap days the rule abstains more, so level CE is high while ΔCE shrinks to 0.129—complementary to the thin-world valve test, not a claim that thin days are more profitable to trade.

Table 10: Content probe by archive density (OOS). ΔCE uses bootstrap 0.334 on All OOS.

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

*Block-bootstrap ΔCE ; raw mean gap $0.130 - (-0.206) = 0.336$.

**Figure 4: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.**

5.5 Discussion

What carries the paper. (i) *The data end is legible*: grounding recovers a PIT-checkable market view (RQ1)—the primary product claim. (ii) *Thin views must not drive action*: soft text fails; the hard valve works across vendors (RQ2). (iii) *Fields are nonempty* under a no-LLM probe and an open-slice content lower bound (RQ3). Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we contribute the data/observation end those systems assume—not a competing decision or execution layer.

Why grounding is the right primary metric. Trading PnL confounds observation quality with policy skill and belongs to the decision end. Grounding asks the data-end question: can the consumer state what is ready, missing, and signed, in a form that can be audited? Raw shows that without the bundle, public LLMs cannot reconstruct that view; Compiled shows they can. The valve then answers the next data-end question: when the disclosed world is thin, will soft judgment stop action? Empirically, no.

Design rationale. (i) Ship a world bundle, not a feature dump. (ii) Previous-close PIT for recomputable `decision_asof`. (iii) Four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. Frozen prompts, temperature 0, IS/OOS cut, and seeds make the four-arm ranking reproducible; an anonymized harness ships on acceptance.

6 Conclusion

End-to-end crypto quant for public LLMs needs a reliable data end before decision or execution. We contribute that observation

layer: compile asynchronous evidence into a consumable PIT world bundle, and force abstention when the view is too thin. Live, grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. Raw collapse; case-level rationales show the same gap; the hard valve abstains 1.0 on thin days where soft judgment does not; vintaged tilts and open-slice fields give a nonempty content lower bound. Build the quant data end first: compile so models can see; refuse when they must not act.

References

- [1] Chanyeol Choi, Jihoon Kwon, Alejandro Lopez-Lira, Chaewoon Kim, Minjae Kim, Juneha Hwang, Jaeseon Ha, Hojun Choi, Suyleol Yun, Yongjin Kim, and Yongjae Lee. 2025. FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 632–637. doi:10.1145/3768292.3770362
- [2] Dean Croushore and Tom Stark. 2001. A Real-Time Data Set for Macroeconomists. *Journal of Econometrics* 105, 1 (2001), 111–130.
- [3] Yitong Duan, Chuheng Zhang, and Jian Li. 2025. FactorMAD: A Multi-Agent Debate Framework Based on Large Language Models for Interpretable Stock Alpha Factor Mining. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 605–613.
- [4] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research* 11 (2010), 1605–1641.
- [5] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates.
- [6] David Ha and Jürgen Schmidhuber. 2018. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates.
- [7] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. 2020. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*. OpenReview.net.
- [8] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. ... and the Cross-Section of Expected Returns. *Review of Financial Studies* 29, 1 (2016), 5–68.
- [9] Yann LeCun. 2022. A Path Towards Autonomous Machine Intelligence. *OpenReview preprint* (2022).
- [10] Yang Li, Yangyang Yu, Haohang Li, Zhiyang Chen, and Khaldoun Khashanah. 2023. TradingGPT: Multi-Agent System with Layered Memory and Distinct Characters for Enhanced Financial Trading Performance. *arXiv preprint arXiv:2309.03736* (2023).
- [11] Yukun Liu, Aleh Tsyvinski, and Xi Wu. 2022. Common Risk Factors in Cryptocurrency. *Journal of Finance* 77, 2 (2022), 1133–1177.
- [12] Alejandro Lopez-Lira and Yuehua Tang. 2023. Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. *arXiv preprint arXiv:2304.07619* (2023).
- [13] Igor Makarov and Antoinette Schoar. 2020. Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics* 135, 2 (2020), 293–319.
- [14] Dimitris N. Politis and Joseph P. Romano. 1994. The Stationary Bootstrap. *J. Amer. Statist. Assoc.* 89, 428 (1994), 1303–1313.
- [15] Yiqing Shen, Jingshu Zhang, Feng Chen, Kaiyuan Yan, and Hongguang Li. 2025. FinSearch: A Temporal-Aware Search Agent Framework for Real-Time Financial Information Retrieval with Large Language Models. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM. doi:10.1145/3768292.3770382
- [16] Rui Sun, Zuo Bai, Wentao Zhang, Yuxiang Zhang, Li Zhao, Shan Sun, and Zhengwen Qiu. 2025. FinResearchBench: A Logic Tree based Agent-as-a-Judge Evaluation Framework for Financial Research Agents. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 656–664. doi:10.1145/3768292.3770364
- [17] Halbert White. 2000. A Reality Check for Data Snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [18] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-Source Financial Large Language Models. In *FinLLM Symposium at IJCAI*.
- [19] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*. OpenReview.net.
- [20] Yangyang Yu, Haohang Li, Zhiyang Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. 2024. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. *Proceedings of the AAAI Symposium Series* 3, 1 (2024), 595–597.
- [21] Wentao Zhang, Lingxuan Zhao, Hao Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, et al. 2024. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM.